

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Wednesday 24 May 2023

Afternoon (Time: 40 minutes)

Paper reference **31617H/1P**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Physics

SECTION C: WAVES IN COMMUNICATION

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises of three papers worth 30 marks each:
 - Section A: Structures and Functions of Cells and Tissues (Biology)
 - Section B: Periodicity and Properties of Elements (Chemistry)
 - Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The formulae sheet can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☐.

- 1 (a) Figure 1 shows a mass at rest hanging from a spring.

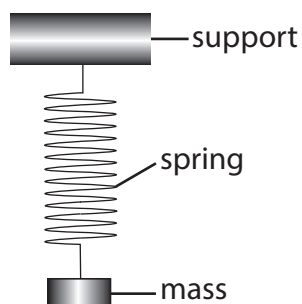


Figure 1

A force is applied to the mass.

The mass is lifted and released so that it oscillates about its rest position.

The displacement of the mass is measured from its rest position.

Figure 2 shows how the displacement of the mass varies with time.

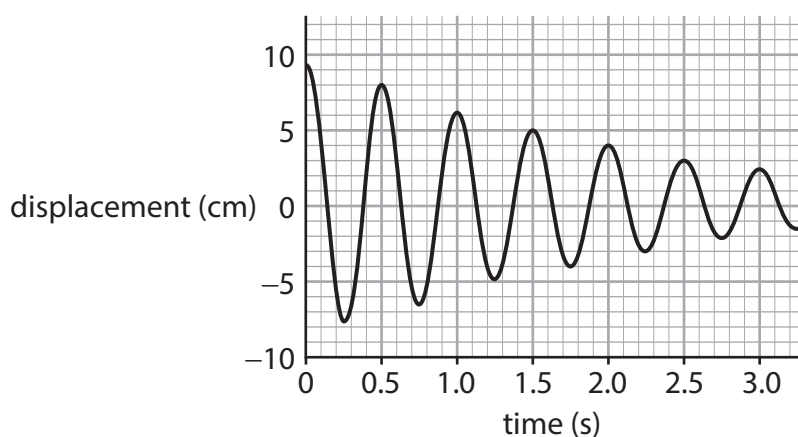


Figure 2

- (i) Give the amplitude of the oscillation at the time 2.0 s.

(1)

amplitude at 2.0 s = cm

- (ii) Give the periodic time (T) for one complete oscillation.

(1)

time for one oscillation (T) = s



(iii) Calculate the frequency (f) of the oscillations.

(2)

Use the equation: $f = \frac{1}{T}$

Show your working.

frequency of oscillations (f) = Hz

(b) Figure 3 shows two waves which are to be superposed.

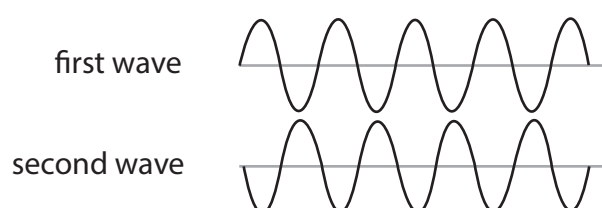


Figure 3

Which row A, B, C or D of the table correctly identifies the type of superposition and the phase difference between the waves.

(1)

		type of superposition	phase difference between waves
☐	A	constructive	180°
☐	B	constructive	360°
☐	C	destructive	180°
☐	D	destructive	360°

(Total for Question 1 = 5 marks)



2 Longitudinal and transverse are two types of wave.

(a) Figure 4 shows a longitudinal wave produced on a long spring.

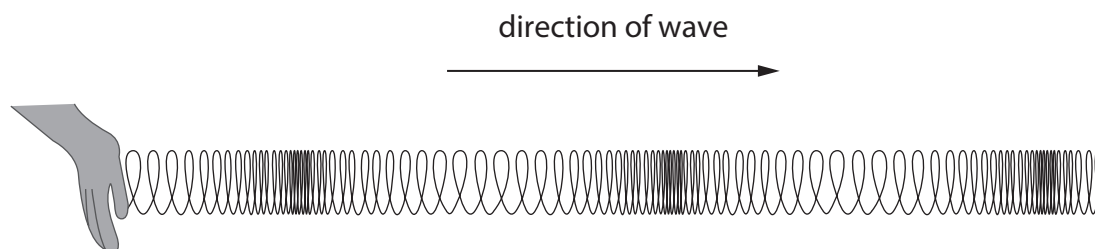


Figure 4

Complete Sentence 1 in the box to describe the longitudinal wave in Figure 4.

The longitudinal wave travels along the spring as a series

of and

Sentence 1

(2)

(b) Infrared waves are an example of transverse waves.

Give **one** use of infrared waves in communication.

(1)



(c) Figure 5 shows a transverse wave being produced using a long spring.

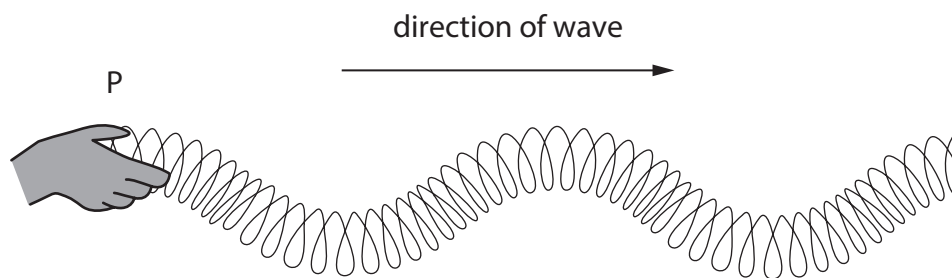


Figure 5

Describe how the movement of the spring at P produces a transverse wave.

(2)

(Total for Question 2 = 5 marks)



- 3 Figure 6 shows a string under tension. An oscillator is vibrating the string to produce a stationary wave.

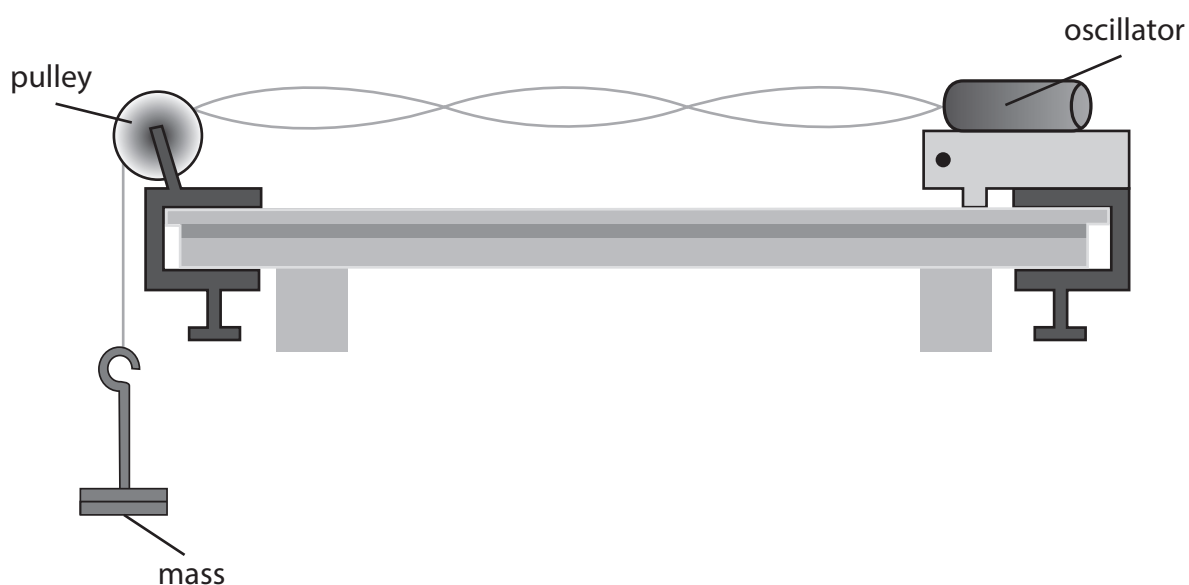


Figure 6

- (a) Describe how a stationary wave is produced in figure 6 using the equipment shown.

(3)

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(b) Which harmonic of the wave is being produced on the string in Figure 6?

(1)

- ☐ **A** first harmonic
- ☐ **B** second harmonic
- ☐ **C** third harmonic
- ☐ **D** fourth harmonic

(c) The string has a mass per unit length (μ) of $6.6 \times 10^{-4} \text{ kg m}^{-1}$.

The velocity (v) of the wave on the string is 32.1 m s^{-1} .

Calculate the tension (T) in the string.

(4)

Use the equation: $v = \sqrt{\frac{T}{\mu}}$

Show your working.

tension (T) in the spring = N

(Total for Question 3 = 8 marks)



- 4 (a) Which row shows regions of the electromagnetic spectrum in order of increasing frequency?

(1)

- ☐ **A** infrared, radio wave, microwave
- ☐ **B** microwave, radio wave, infrared
- ☐ **C** radio wave, infrared, microwave
- ☐ **D** radio wave, microwave, infrared

- (b) Mobile phones use electromagnetic waves for communication.

Describe how a signal is transmitted from one mobile phone to another mobile phone.

(3)

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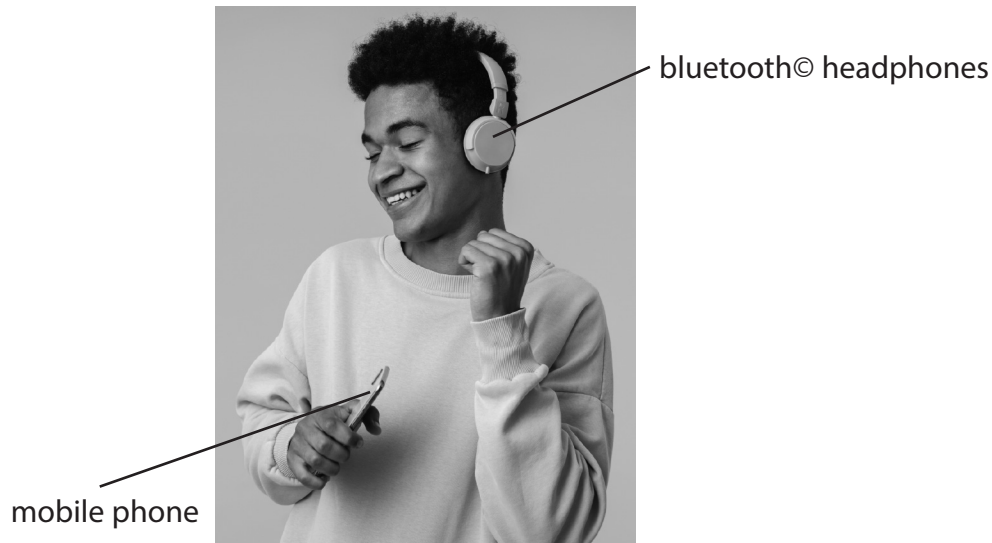
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(c) A student uses Bluetooth® headphones to listen to music.



(Source ©PAL)

Figure 7

Explain why the Bluetooth® headphones must be near to the mobile phone so that the student can listen to the music.

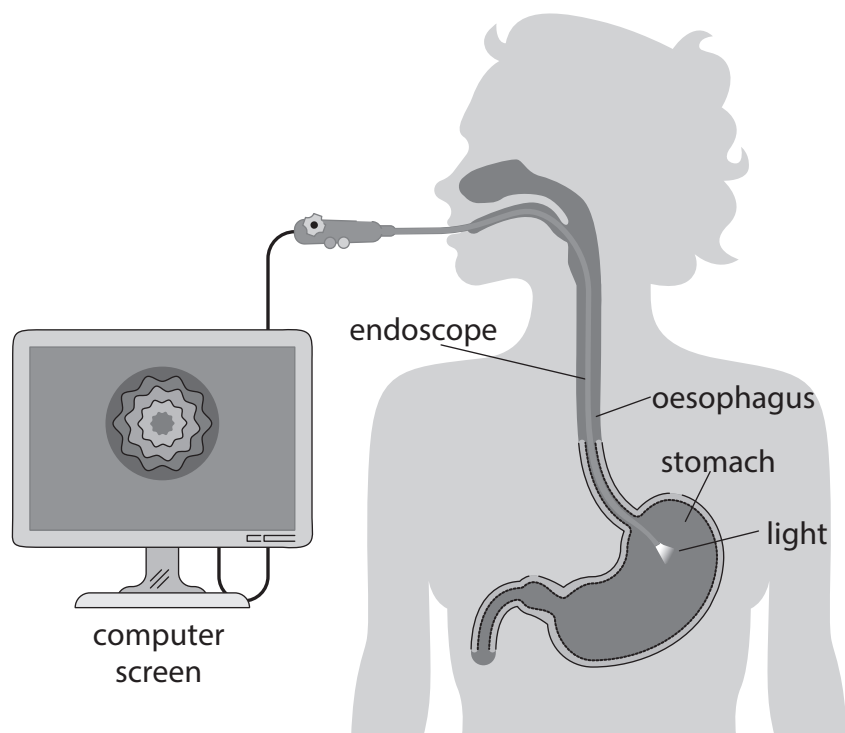
(2)

(Total for Question 4 = 6 marks)



- 5 Figure 8 shows an endoscope being used to observe the inside of the stomach of a patient.

Endoscopes use optical fibres.



(Source ©PAL)

Figure 8

Explain how the optical fibres in an endoscope are used to produce an image on the computer screen.

(6)

You may add a diagram to support your answer.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 5 = 6 marks)

TOTAL FOR SECTION C = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



P 7 4 4 9 6 A 0 1 1 1 2

Formulae sheet

Wave speed

$$v = f\lambda$$

Speed of a transverse wave on a string

$$v = \sqrt{\frac{T}{\mu}}$$

Refractive index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

Critical angle

$$\sin C = \frac{1}{n}$$

Inverse square law in relation to the intensity of a wave

$$I = \frac{k}{r^2}$$

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

